

Report R27 — the evolution operator’s Schmidt rank

What the literature establishes, and why a linear-dimensional module forces linear growth

Tier 1 analysis

2026-08-22

Abstract

Rule 156 has an operator Schmidt rank $\chi(t)$ of its *evolution operator* that grows linearly in t , in a clean, disorder-free, translation-invariant Floquet circuit. This report establishes what the literature says about that observable and supplies a mechanism. §1 is an audited pass over the twelve references supplied plus eleven found by search, sorted by relevance, each with the finding that bears on our claim. **The baseline is Dubail’s:** the OSEE of $U(t) = e^{-iHt}$ grows *linearly*, hence χ *exponentially*, “unless the Hamiltonian is in a localized phase”. Zhou and Luitz classify the three observed growth laws for $U(t)$ — linear, power law, logarithmic — and both sub-linear classes occur only with disorder. Every disorder-free sub-linear result we could find, including Alba–Dubail–Medenjak for Rule 54 and the very recent Tan–Prosen clean non-integrable circuit, is about *local operators in the Heisenberg picture*, which Dubail explicitly separates from the propagator. We found no disorder-free system with sub-linear evolution-operator OSEE. §2 then derives the mechanism. The key quantity is not integrability but the **number of defects that can cross the cut**: if the sector containing the cut is spanned by configurations of p domain walls with p bounded, then $\chi(t) = O(t^p)$, and if p is extensive χ grows exponentially. The kink module of W_{156}/W_{198} has $p = 1$, giving $\chi(t) = O(t)$; the hard-core module of W_{201}/W_{108} has extensive p , giving exponential growth. Fragmentation into modules of polynomial dimension is therefore a disorder-free mechanism for the localization signature.

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1 Protocol: the literature, sorted by relevance

Two objects must be kept apart throughout, because the literature treats them very differently and conflating them is the trap this project already fell into once:

- **(L)** the operator entanglement of a *local* operator $O(t) = U^\dagger O U$ in the Heisenberg picture;
- **(U)** the operator entanglement — equivalently the Schmidt rank of the Choi vector, equivalently the MPO bond dimension — of the *evolution operator* $U(t)$ itself.

Our measurement is **(U)**. Sub-linear results for **(L)** do not bear on it.

1.1 Tier 1: directly on the claim

reference	object	finding that bears on our claim
Dubail, J. Phys. A 50 , 234001 (2017)	(U)	The decisive baseline. “The OSEE of the evolution operator $U(t) = e^{-iHt}$ increases linearly with t , unless the Hamiltonian is in a localized phase.” Explicitly contrasts this with local operators, whose OSEE “grows sublinearly (perhaps logarithmically)”. Linear OSEE $\Rightarrow \chi$ exponential. <i>Localization is named as the only exception.</i>
Zhou & Luitz, PRB 95 , 094206 (2017)	(U)	Classifies the growth of operator EE of the time evolution operator across chaotic, MBL and Floquet systems: linear, power-law and logarithmic growth are all observed, before extensive saturation. Crucially the two sub-linear classes are the <i>random-field</i> Heisenberg model — power law at weak disorder, logarithmic at strong disorder. The most chaotic Floquet model saturates at the Page value.
Thakur, Pain & Roy, PRB 111 , 174206 (2025)	(U)	Operator entanglement of the time-evolution operator in the MBL phase, showing a <i>logarithmic light cone</i> , derived from eigenstate correlations without invoking ℓ -bits. This is the closest published phenomenology to ours — same object, same functional form — but obtained <i>from disorder</i> .
Prosen & Žnidarič, PRE 75 , 015202 (2007)	(L)/state	The origin of the exponential-versus-polynomial dichotomy for MPO bond dimension: $D_\epsilon(t) \propto e^{ct}$ in quantum chaotic systems, polynomial when integrable. <i>But the evolved objects are observables and thermal states, not the propagator</i> , so this does not license “integrable $\Rightarrow$ polynomial χ ” for $U(t)$.

1.2 Tier 2: the disorder-free sub-linear results, all for local operators

reference	object	finding
Alba, Dubail & Medenjak, PRL 122 , 250603 (2019)	(L)	Logarithmic upper bound on operator entanglement for <i>all local operators</i> in the Rule 54 chain, and evidence that log growth is generic for interacting integrable systems. A cellular automaton, so tempting — but the wrong object.

reference	object	finding
Tan & Prosen, (L) arXiv:2603.19363 (2026)		“Logarithmic growth of operator entanglement in a <i>clean non-integrable</i> circuit”. Semi-ergodic dual-unitary brickwork; mechanism is that the Heisenberg evolution of a single traceless single-site operator lies in a restricted subspace. The nearest disorder-free precedent in spirit — log growth from a restricted subspace rather than from integrability or disorder — but again for a local operator, not $U(t)$.
Prosen & Pizorn, PRA 76 , 032316 (2007)	(L)	Transverse Ising chain: OSEE of local operators <i>saturates</i> for finite Majorana index and grows <i>logarithmically</i> for infinite index. Free fermions; local operators.
Pal & Lakshminarayan, PRB 98 , 174304 (2018)	(U)	Entangling power of time-evolution operators, integrable versus non-integrable. The integrable kicked transverse-field Ising chain still shows <i>ballistic</i> growth of operator von Neumann entropy; its entangling power saturates lower than the non-integrable case but the growth is not sub-linear. <i>Integrability alone does not buy sub-linear</i> $\chi(t)$.

1.3 Tier 3: framework, and the closest methodological neighbour

reference	object	finding
Merbis & Bakker, ICCS 2024, arXiv:2406.04895	(U), classical	The closest method: builds the MPO of a cellular automaton’s transition matrix and classifies elementary CA by operator-entanglement growth. Types I (converges to a constant, e.g. their rules 136, 108), II (rises then falls, peak at $t \sim L$), III (linear growth to an L -dependent plateau, chaotic-like). Generic bond dimension quoted as 4^t , i.e. $S(t) = 2t$. <i>No logarithmic class appears</i> . Their object is the classical transition matrix of a deterministic CA, not a quantum evolution operator with Hadamards, so the rule numbers are not comparable to ours despite the coincidence.
Zanardi, Zalka & Faoro, PRA 62 , 030301 (2000); Zanardi, PRA 63 , 040304 (2001)	(U)	Where the object comes from: entanglement of a unitary regarded as a vector in Hilbert–Schmidt space, and the entangling power. Definitional; sets the measure we and everyone else use.
Jonay, Huse & Nahum, arXiv:1803.00089 (2018)	(L)/(U)	The membrane / line-tension coarse-graining. Predicts <i>linear</i> growth for generic chaotic dynamics with rate set by the surface tension $\mathcal{E}(v)$. Our result sits outside this picture, which is the point.

reference	object	finding
Hahn, Luitz & Chalker, PRX 14 , 031029 (2024)	(U)	Eigenstate correlations of the time-evolution operator beyond ETH; uses Floquet models without conserved densities to isolate scrambling. Provides the eigenstate-correlation language in which Thakur–Pain–Roy then derive the MBL log light cone.
Bandyopadhyay & Lakshminarayan, arXiv:quant-ph/0504052 (2005)	(U)	Entangling power of quantum chaotic evolutions; early evidence that chaotic evolutions approach the random-matrix value.
Musz, Kuś & Życzkowski, PRA 87 , 022111 (2013)	(U)	Geometry of unitary gates, perfect entanglers, unistochastic maps. Fixes what “maximal” means for the measure.
arXiv:2603.05656 (2026), <i>Classical simulability from operator entanglement scaling</i>	(L)	Simulability classification by the scaling exponent α : $\alpha \geq 1$ blocks efficient MPO approximation, $\alpha < 1$ permits it over restricted state ensembles. Local-operator entanglement.

1.4 Verdict of the pass

1. For the *evolution operator*, the established law is linear OSEE, i.e. **exponential** $\chi(t)$, with exactly one documented exception: **localization**, which requires disorder (Dubail; Zhou–Luitz; Thakur–Pain–Roy).
2. Integrability is *not* an exception for this object. Prosen–Žnidarič’s polynomial result is for observables and states; Pal–Lakshminarayan find ballistic operator-entropy growth even for an integrable kicked Ising chain.
3. Every disorder-free sub-linear result we found — Alba–Dubail–Medenjak, Prosen–Pižorn, Tan–Prosen — is for *local operators*, which Dubail explicitly separates from the propagator.
4. We found **no** report of sub-linear evolution-operator Schmidt-rank growth in a clean, disorder-free system. A linear $\chi(t)$ for $U(t)$ in a translation-invariant Floquet circuit is, on this evidence, unoccupied ground.

Remark 1. The nearest conceptual precedent is Tan–Prosen: log growth in a clean non-integrable circuit because the evolved operator is confined to a *restricted subspace*. That is the same shape of mechanism as ours — confinement to a small module rather than integrability or disorder — applied to the other object. It should be cited as the closest relative and distinguished on the object.

2 The mechanism: defect counting bounds the Schmidt rank

We now give the argument that a module of linear dimension forces linear $\chi(t)$. The statement turns out not to be about dimension as such, but about *how many defects can cross the cut*.

2.1 Setup

Let U be one period of a brick-wall circuit with strict light-cone velocity v (sites per period). Fix a cut at bond x and let $\chi_x(t)$ be the operator Schmidt rank of $U(t) = U^t$ across it —

equivalently the rank of the Choi vector $|U(t)\rangle\rangle$ under the bipartition, equivalently the MPO bond dimension at that bond.

The generic bound is the light cone: $U(t)$ acts as the identity outside a causal diamond of width $2vt$ about the cut, so $\chi_x(t) \leq 4^{2vt}$. Everything in §1 that grows exponentially is saturating a bound of this form. To do better one needs the dynamics to explore only a small part of the operator space on the diamond.

2.2 The kink chain

Take W_{156} . Its sectors are wall-free gaps, and inside a gap the state space is spanned by the *kink states*

$$|k\rangle = \underbrace{1 \cdots 1}_k \underbrace{0 \cdots 0}_{l-k}, \quad k = 0, \dots, l,$$

of dimension $l+1$ — one degree of freedom, the position of a single domain wall. Restricted to the gap,

$$U(t)|_{\text{gap}} = \sum_{k,k'} A_{kk'}(t) |k\rangle\langle k'|,$$

a *single-particle propagator*. Two facts make the Schmidt rank collapse.

Every kink state is a product across the cut. Cutting at x ,

$$|k\rangle = |L_k\rangle \otimes |R_k\rangle, \quad \begin{cases} L_k = 1^k 0^{x-k}, & R_k = 0^{l-x} & k \leq x, \\ L_k = 1^x, & R_k = 1^{k-x} 0^{l-k} & k > x. \end{cases}$$

So each term $|k\rangle\langle k'|$ is already a product operator $\ell_{kk'} \otimes r_{kk'}$ with $\ell_{kk'} = |L_k\rangle\langle L_{k'}|$ and $r_{kk'} = |R_k\rangle\langle R_{k'}|$. No entanglement is generated by the basis itself.

Three of the four index sectors are rank one. Split the sum by the side of the cut:

- $k, k' \leq x$: then $R_k = R_{k'} = 0^{l-x}$, so *every* term shares the same right factor. All of these together contribute Schmidt rank ≤ 1 .
- $k, k' > x$: then $L_k = L_{k'} = 1^x$, the same left factor. Rank ≤ 1 .
- $k \leq x < k'$ and $k' \leq x < k$: the crossing sectors. Here the left factor is $|L_k\rangle\langle 1^x|$, labelled by k alone, and the right factor is $|0^{l-x}\rangle\langle R_{k'}|$, labelled by k' alone. The contribution is the rank of the coefficient block $A_{kk'}$ restricted to that range.

Causality restricts the crossing blocks: $A_{kk'}(t) = 0$ unless $|k - k'| \leq 2vt$, and a term contributes across the cut only if the kink is within the light cone of x , i.e. $|k - x|, |k' - x| \lesssim vt$. Each crossing block is therefore at most $(vt) \times (vt)$, of rank at most vt . Hence

$$\boxed{\chi_x(t) \leq 2 + 2vt = O(t).} \tag{1}$$

2.3 The general statement

Nothing above used δ , the Hadamard, or integrability. What was used is that the sector is labelled by the position of *one* defect.

Proposition 1 (Defect counting). *Suppose U preserves a sector whose basis states are labelled by the positions of p defects on the chain, each defect state being a product state across any cut, and suppose U has strict light-cone velocity v . Then the operator Schmidt rank of $U(t)$ across a cut obeys*

$$\chi_x(t) = O((vt)^p).$$

If instead the number of defects is extensive in the region size, the bound degrades to the light-cone bound and $\chi_x(t)$ may grow exponentially.

Sketch. As above: a configuration is fixed by the p defect positions, so a basis operator $|c\rangle\langle c'|$ factorises across the cut with left factor determined by the defect positions to the left and right factor by those to the right. Terms in which no defect lies within the light cone of the cut share a common factor on one side and contribute rank 1. Terms with $j \leq p$ defects inside the light cone are labelled by at most $(2vt)^j$ positions on each side, so the corresponding coefficient block has rank at most $(2vt)^j$. Summing over $j \leq p$ gives $O((vt)^p)$. $\square$

2.4 The two families

rules	gap module	dimension	defects p	predicted $\chi(t)$
W_{156}, W_{198}	kink chain $1^a 0^b$	$l + 1$, linear	1	$O(t)$, linear
W_{201}, W_{108}	hard-core chain	$\sim \varphi^l$, exponential	extensive	exponential

This reproduces both measured behaviours: linear $\chi(t)$ for rule 156, exponential for 201/108. It also explains why the split is so sharp despite the two families differing by a single letter of the gate word: the V -context decides the wall word, the wall word decides whether a gap admits an ascent, and that decides whether the gap carries one defect or extensively many.

Remark 2 (What the argument does and does not give). It gives linearity, and the identification of the exponent with the defect number. It does *not* give the measured slope 6; that requires counting the wall-pattern boundary conditions at the cut and the brick-wall velocity together, and is not attempted here. It is an upper bound, so it does not by itself prove the growth is not slower. And it assumes the sector basis is product across the cut, which holds for kink and hard-core configurations but must be checked for any other module.

Remark 3 (Why this is the localization signature without localization). In an MBL system the log light cone comes from an emergent set of quasi-local integrals of motion which confine information transport. Here the confinement is kinematic: the sector simply has too few configurations within the light cone for the Schmidt rank to grow. Fragmentation supplies, without disorder, the thing localization supplies with it — a bound on how many distinguishable histories can cross a cut in time t . That is the sense in which Proposition 1 is the mechanism the literature pass says is missing.

3 What would settle it

1. $\chi(t)$ **for** $W_{198}, W_{201}, W_{108}$. The linear-versus-exponential contrast is the whole claim, and only W_{156} has been measured in this project. The unitary path is the cheap one and the existing code already takes the rule as an argument.
2. **A lower bound.** Proposition 1 bounds χ from above. Showing $\chi = \Theta(t)$ rather than $O(t)$ needs the crossing block of $A_{kk'}$ to have full rank, which for a single-particle propagator on a line is plausible but unproved here.
3. **The slope.** 6 per period is exact in the measured window and is presumably a small combinatorial count; deriving it would considerably strengthen the result.

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